Fast integrators with a vertically implicit step: the WRF forward–backward scheme and Verlet EI, Cartesian and terrain-following

MultiRate.jl

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Abstract

We describe the acoustic small step of WRF (forward–backward, vertically implicit, off-centred) and the linearly implicit Verlet step "Version EI" of the note *Split-explicit with implicit treatment of part of the fast waves* (Imex.pdf, 2024), first in Cartesian coordinates and then on a terrain-following grid, where the contravariant vertical mass flux Ω is the unknown of the column system and all vertical pressure derivatives are implicit. The pressure is linearized throughout. Both schemes have the same two options, off-centring and explicit extrapolation (divergence damping); for equal options they are conjugate. The schemes are implemented in the frozen MIS split with the W system and an explicit metric term (Section 8) and compared on the Agnesi mountain wave (Section 9).

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1 Introduction

Split-explicit methods (MIS, RK3) integrate the fast acoustic–gravity part with small steps $\Delta\tau = h$. On grids with $\Delta z \ll \Delta x$ the vertical acoustic Courant number $c_s\Delta\tau/\Delta z$ is far above one. The vertical part has to be implicit; the horizontal part stays explicit (HEVI). Three ingredients decide the properties of such a small step:

1. the explicit horizontal integrator (forward–backward, Störmer–Verlet);
2. the implicit vertical integrator (Crank–Nicolson, off-centred) and the variable whose column system is solved (W, or Ω on terrain);
3. which pressure derivatives are implicit on terrain.

Notation: ϑ denotes the thermodynamic variables, $\vartheta = (\rho, \Theta)$ with $\Theta = \rho\theta$ (in the extended forms also p or π, in the moist form ρq_v, ρq_c); $U = \rho u$ and $W = \rho w$ are the momenta; θ_f is the frozen potential temperature at the faces; $h = \Delta\tau$.

2 The fast system with linearized pressure

2.1 Linearized pressure

In the fast part the pressure enters only through its gradient. It is linearized about the frozen state of the stage,

$$p = p_f + P\,(\Theta - \Theta_f),$$
$$P = \Big(\frac{\partial p}{\partial \Theta}\Big)_f = \Big(\frac{\gamma p}{\Theta}\Big)_f,$$

with P frozen in space and time within a stage. The gradient of $p_f - P\Theta_f$ is part of the constant forcing, so in the fast operator $p = P\Theta$. With the pressure itself as an additional variable the pressure is linear by construction (Θ replaced by p, $P = 1$); with the Exner pressure the gradient is $c_p\Theta_f\nabla\pi$ (P replaced by $c_p\Theta_f$). The algebra below holds for all three.

2.2 Cartesian coordinates

With frozen face values θ_f the fast system is

$$\begin{aligned}
\rho_t &= -\partial_x U - \partial_z W,\\
\Theta_t &= -\partial_x(\theta_f U) - \partial_z(\theta_f W),\\
U_t &= -\partial_x p,\\
W_t &= -\partial_z p - g\rho,\\
p &= P\,\Theta.
\end{aligned}$$

In block form, with the horizontal and the vertical divergence D_h, D_v and the pressure forces $G_x = -\partial_x P$, $B_v = -\partial_z P - g$ (acting on Θ and ρ):

$$\begin{aligned}
\vartheta_t &= D_h U + D_v W + f_\vartheta,\\
U_t &= G_x \vartheta + f_U,\\
W_t &= B_v \vartheta + f_W.
\end{aligned}$$

f is the constant forcing of the stage (slow tendencies, linearization remainder). The vertical part (D_v, B_v) is a closed system in each column.

2.3 Terrain-following coordinates and the contravariant mass flux

Gal-Chen coordinate $z = z_s(x) + \eta J(x)$, $J = 1 - z_s/H$, metric $z_x = \partial_x z|_\eta = z_s'(1 - \eta/H)$. The contravariant vertical mass flux through the η-surfaces is

$$\Omega = W - z_x U.$$

The fast system in flux form reads

$$\begin{aligned}
J\rho_t &= -\partial_x(JU) - \partial_\eta \Omega,\\
J\Theta_t &= -\partial_x(J\theta_f U) - \partial_\eta(\theta_f \Omega),\\
U_t &= -\frac{1}{J}\big[\partial_x(Jp) - \partial_\eta(z_x p)\big],\\
W_t &= -\frac{1}{J}\partial_\eta p - g\rho.
\end{aligned}$$

Write the horizontal pressure force and the vertical pressure derivative separately,

$$G_x p = -\frac{1}{J}\big[\partial_x(Jp) - p\,\partial_\eta z_x\big],$$

$$G_\eta p = \frac{1}{J}\,\partial_\eta p.$$

Then $U_t = G_x p + z_x G_\eta p$, and from $\Omega_t = W_t - z_x U_t$

$$\begin{aligned}
U_t &= G_x p + z_x\, G_\eta p,\\
\Omega_t &= -(1 + z_x^2)\, G_\eta p - g\rho - z_x\, G_x p.
\end{aligned}$$

All vertical pressure derivatives $G_\eta p$ — in the U equation through the metric term and in the Ω equation with the factor $1 + z_x^2$ — and gravity are treated implicitly; the horizontal pressure force $G_x p$ (in U and in Ω) and the horizontal divergence $\partial_x(JU)$ explicitly. The thermodynamic equations contain the momenta in the vertical only through Ω, so (ϑ, Ω) is a closed column system for the implicit part. U and W remain the prognostic variables; Ω is the variable of the column solve, and U, W are computed from the new pressure afterwards (Section 5). For $z_x = 0$, $J = 1$ this is the Cartesian system with $\Omega = W$.

Why the metric term has to be implicit on steep terrain. Constant slope $s = z_x$, no gravity, vertical structure only:

$$\Omega_t = -(1+s^2)\,\frac{\partial_\eta p}{\rho_0},$$

the vertical acoustic mode of the terrain-following coordinate has the frequency $c_s k_\eta \sqrt{1+s^2}$. If the metric term $z_x G_\eta p$ in the U equation is explicit (as in WRF and in the present code), the part s^2 of the restoring force passes through the explicit U. A heuristic bound from the largest vertical wavenumber $k_\eta \approx 2/(J\Delta\eta)$ is

$$\frac{c_s\,\Delta\tau\,|z_x|}{J\Delta\eta} \lesssim 1.$$

(An estimate, not a stability analysis of the coupled step.) Agnesi case: $350 \cdot 4.3 \cdot 0.026/200 \approx 0.2$; steep terrain, $z_x = 0.5$, $\Delta z = 50\,\mathrm{m}$: ≈ 15.

3 The WRF idea in Cartesian coordinates

3.1 WRF

WRF integrates the acoustic modes in the small steps of its RK3 (Wicker and Skamarock 2002), in a terrain-following hydrostatic-pressure (mass) coordinate with column mass μ, geopotential ϕ and perturbation variables about the state of the large step (Klemp, Skamarock and Dudhia 2007; Skamarock et al. 2019). One small step: (1) horizontal momentum forward with the divergence-damped pressure; (2) column mass and Ω from the new horizontal fluxes (Ω is diagnostic in the mass coordinate); (3) Θ with the new fluxes; (4) W and ϕ vertically implicit, off-centred, eliminated to a tridiagonal system for W. The idea, stripped of the mass coordinate, is the following forward–backward step in Cartesian coordinates.

3.2 The step

With the off-centring $\beta = (1+\varepsilon)/2$ and the weighted average

$$a^\beta = (1-\beta)\,a^n + \beta\,a^{n+1},$$

one small step $n \to n+1$ is

$$\tilde{p} = p^n + \gamma_d\,(p^n - p^{n-1}), \tag{1}$$

$$U^{n+1} = U^n - h\,\partial_x \tilde{p} + h f_U, \tag{2}$$

$$\rho^{n+1} = \rho^n - h\,\partial_x U^{n+1} - h\,\partial_z W^\beta + h f_\rho, \tag{3}$$

$$\Theta^{n+1} = \Theta^n - h\,\partial_x(\theta_f U^{n+1}) - h\,\partial_z(\theta_f W^\beta) + h f_\Theta, \tag{4}$$

$$W^{n+1} = W^n - h\,\partial_z p^\beta - h\,g\rho^\beta + h f_W. \tag{5}$$

The horizontal part is forward–backward: U forward with the old (extrapolated) pressure, ϑ backward with the new U. The vertical part is the β-weighted Crank–Nicolson step of (ϑ, W) with U^{n+1} fixed. Two options, both zero by default:

- off-centring ε (WRF: `epssm` $= 0.1$): damps the vertically propagating acoustic modes;
- explicit extrapolation γ_d (WRF: `smdiv` $= 0.1$, divergence damping): damps the horizontally propagating acoustic modes. In the first small step of a stage $p^{n-1} = p^n$.

3.3 Elimination: the system for W

After (2) U^{n+1} is known. Collect the known parts of (3)–(4),

$$\hat\rho = \rho^n - h\,\partial_x U^{n+1} - (1-\beta)h\,\partial_z W^n + h f_\rho,$$
$$\hat\Theta = \Theta^n - h\,\partial_x(\theta_f U^{n+1}) - (1-\beta)h\,\partial_z(\theta_f W^n) + h f_\Theta,$$

so that

$$\rho^{n+1} = \hat\rho - \beta h\,\partial_z W^{n+1},$$
$$\Theta^{n+1} = \hat\Theta - \beta h\,\partial_z(\theta_f W^{n+1}),$$
$$p^{n+1} = P\,\Theta^{n+1}.$$

Inserted into (5):

$$W^{n+1} - \beta^2 h^2\,\partial_z\big(P\,\partial_z(\theta_f W^{n+1})\big) - \beta^2 h^2\,g\,\partial_z W^{n+1} = R_W,$$

$$R_W = W^n - h\,\partial_z\big[(1-\beta)p^n + \beta P\hat\Theta\big] - hg\big[(1-\beta)\rho^n + \beta\hat\rho\big] + h f_W.$$

The operator is a discrete Helmholtz operator in z: on the C-grid a tridiagonal matrix per column (Section 6.1). Afterwards ρ^{n+1}, Θ^{n+1} follow from the two lines above (back substitution). For $\beta = \frac{1}{2}$ the matrix is $I - \frac{h^2}{4}(\partial_z P \partial_z \theta_f + g\partial_z)$.

4 Verlet EI in Cartesian coordinates

4.1 The step

In the Imex note the momentum operator is split into an explicit part (here the horizontal pressure force) and an implicit part (the vertical pressure force and gravity); the divergence is not split. With the same options as WRF, $\beta = (1+\varepsilon)/2$, one small step is

$$\vartheta^{n+1/2} = \vartheta^n + (1-\beta)h\,\big[D_h U^n + D_v W^n + f_\vartheta\big], \tag{6}$$
$$W' = W^n + (1-\beta)h\,\big[B_v\vartheta^n + f_W\big], \tag{7}$$
$$U^{n+1} = U^n + h\,G_x\tilde\vartheta^{\,n+1/2} + h f_U, \tag{8}$$
$$\vartheta^{n+1} = \vartheta^{n+1/2} + \beta h\,\big[D_h U^{n+1} + D_v W^{n+1} + f_\vartheta\big], \tag{9}$$
$$W^{n+1} = W' + \beta h\,\big[B_v\vartheta^{n+1} + f_W\big], \tag{10}$$

with the extrapolation

$$\tilde\vartheta^{\,n+1/2} = \vartheta^{n+1/2} + \gamma_d\big(\vartheta^{n+1/2} - \vartheta^{n-1/2}\big)$$

(no extrapolation in the first small step of a stage). For $\varepsilon = \gamma_d = 0$ this is Version EI of the Imex note: thermodynamic half step explicit in the full divergence, U explicit, W trapezoidal in the vertical pressure force and gravity, thermodynamic half step.

4.2 Elimination: the system for W

(9)–(10) are the implicit part. With

$$\vartheta^* = \vartheta^{n+1/2} + \beta h\,\big[D_h U^{n+1} + f_\vartheta\big]$$

they read $\vartheta^{n+1} = \vartheta^* + \beta h\,D_v W^{n+1}$ and $W^{n+1} = W' + \beta h\,[B_v\vartheta^{n+1} + f_W]$, and

$$W^{n+1} - \beta^2 h^2\,\partial_z\big(P\,\partial_z(\theta_f W^{n+1})\big) - \beta^2 h^2\,g\,\partial_z W^{n+1} = W' + \beta h\,\big[B_v\vartheta^* + f_W\big].$$

The same matrix as in WRF. In the original Version III of the Imex note only the W rows are off-centred (γ instead of β in (7), (10), $\frac{h}{2}$ in (6), (9)); the matrix factor is then $\gamma h^2/2$.

4.3 Properties

- Purely vertical problem ($D_h = G_x = 0$): the β-weighted Crank–Nicolson step of the column; for $\varepsilon = 0$ the trapezoidal rule, unconditionally stable and non-dissipative.
- Purely horizontal problem, $\varepsilon = 0$: Störmer–Verlet (position Verlet), symmetric, second order, stable for $|\omega h| \le 2$.
- For $\varepsilon = 0$ the step is symmetric and of second order; for $\varepsilon > 0$ of first order.
- With the half steps of consecutive small steps combined (Section 6.3) the cost per small step is that of WRF: one evaluation of each part and one column solve.

5 Orography: solve in Ω, all vertical pressure derivatives implicit

5.1 The momentum equations in U and W

On the terrain-following grid (Section 2.3) collect the momenta in $M = (U, W)^T$. With the metric vector

$$b = \begin{pmatrix} -z_x \\ 1 \end{pmatrix}$$

the momentum equations are

$$M_t = \begin{pmatrix} G_x p \\ 0 \end{pmatrix} - b\, G_\eta p - \begin{pmatrix} 0 \\ g\rho \end{pmatrix} + f_M.$$

The first term (horizontal pressure force) is explicit; the second (every vertical pressure derivative: the metric term $z_x G_\eta p$ in U and $-G_\eta p$ in W) and the third (gravity) are implicit. The contravariant mass flux is the metric row applied to M,

$$\Omega = b^T M = W - z_x U,$$

and the thermodynamic equations contain the momenta in the vertical only through Ω:

$$\begin{aligned} J\rho_t &= -\partial_x(JU) - \partial_\eta \Omega, \\ J\Theta_t &= -\partial_x(J\theta_f U) - \partial_\eta(\theta_f \Omega). \end{aligned}$$

The horizontal divergence is explicit. With $p = P\Theta$ and $\beta = (1+\varepsilon)/2$ the splitting is

equation	explicit	implicit
ρ	$-\frac{1}{J}\partial_x(JU)$	$-\frac{1}{J}\partial_\eta \Omega$
Θ	$-\frac{1}{J}\partial_x(J\theta_f U)$	$-\frac{1}{J}\partial_\eta(\theta_f \Omega)$
U	$G_x p$	$z_x\, G_\eta p$
W	—	$-G_\eta p - g\rho$

5.2 The WRF step

1. Explicit part. Extrapolated pressure and explicit momenta:

$$\begin{aligned} \tilde{p} &= p^n + \gamma_d\,(p^n - p^{n-1}), \\ \hat{M} &= M^n + h \begin{pmatrix} G_x \tilde{p} \\ 0 \end{pmatrix} - (1-\beta) h\, b\, G_\eta p^n - (1-\beta) h \begin{pmatrix} 0 \\ g\rho^n \end{pmatrix} + h f_M. \end{aligned}$$

$\hat{M} = (\hat{U}, \hat{W})$ contains the horizontal pressure force and the old-time part of the implicit terms.

2. The implicit step in U and W.

$$\rho^{n+1} = \rho^n - \frac{h}{J}\,\partial_x(J\hat{U}) - \frac{h}{J}\,\partial_\eta\Omega^\beta + hf_\rho, \tag{11}$$

$$\Theta^{n+1} = \Theta^n - \frac{h}{J}\,\partial_x(J\theta_f\hat{U}) - \frac{h}{J}\,\partial_\eta(\theta_f\Omega^\beta) + hf_\Theta, \tag{12}$$

$$M^{n+1} = \hat{M} - \beta h\, b\, G_\eta p^{n+1} - \beta h \begin{pmatrix} 0 \\ g\rho^{n+1} \end{pmatrix}, \tag{13}$$

with $\Omega^\beta = (1-\beta)\Omega^n + \beta\Omega^{n+1}$ and $\Omega = b^T M$. The horizontal mass flux uses $\hat{U}$ (point (b) below).

3. Multiply by the metric row. b^T from the left on (13), with $b^T b = 1 + z_x^2$ and $b^T(0,1)^T = 1$:

$$\Omega^{n+1} = \hat{\Omega} - \beta h\,(1+z_x^2)\,G_\eta p^{n+1} - \beta h\, g\rho^{n+1},$$

$$\hat{\Omega} = b^T\hat{M} = \hat{W} - z_x\hat{U}.$$

Together with (11)–(12) this is a closed system for (ρ, Θ, Ω) in each column.

4. Solve in Ω. With the known parts

$$\hat{\rho} = \rho^n - \frac{h}{J}\,\partial_x(J\hat{U}) - (1-\beta)\frac{h}{J}\,\partial_\eta\Omega^n + hf_\rho,$$
$$\hat{\Theta} = \Theta^n - \frac{h}{J}\,\partial_x(J\theta_f\hat{U}) - (1-\beta)\frac{h}{J}\,\partial_\eta(\theta_f\Omega^n) + hf_\Theta,$$

the thermodynamic variables are

$$\rho^{n+1} = \hat{\rho} - \beta\frac{h}{J}\,\partial_\eta\Omega^{n+1}, \tag{14}$$

$$\Theta^{n+1} = \hat{\Theta} - \beta\frac{h}{J}\,\partial_\eta(\theta_f\Omega^{n+1}), \tag{15}$$

and inserting them into the Ω equation gives the tridiagonal system per column

$$\Omega^{n+1} - \beta^2h^2\,\frac{1+z_x^2}{J}\,\partial_\eta\Big(\frac{P}{J}\,\partial_\eta(\theta_f\Omega^{n+1})\Big) - \beta^2h^2\,\frac{g}{J}\,\partial_\eta\Omega^{n+1} = R_\Omega,$$

$$R_\Omega = \hat{\Omega} - \beta h\,(1+z_x^2)\,G_\eta\big(P\hat{\Theta}\big) - \beta h\, g\hat{\rho},$$

with $\Omega = 0$ at the ground and at the top. Compared with the Cartesian W system: $\partial_z \to J^{-1}\partial_\eta$ and the factor $1 + z_x^2$ on the pressure part.

5. Back substitution, then U and W from the pressure. ρ^{n+1} and Θ^{n+1} from (14)–(15),

$$p^{n+1} = P\,\Theta^{n+1},$$

and then both momenta from (13) with the known p^{n+1} and ρ^{n+1}:

$$U^{n+1} = \hat{U} + \beta h\, z_x\, G_\eta p^{n+1},$$
$$W^{n+1} = \hat{W} - \beta h\, G_\eta p^{n+1} - \beta h\, g\rho^{n+1}.$$

U and W are not recovered from Ω; they are computed from the pressure, and $b^T M^{n+1} = \Omega^{n+1}$ holds because step 3 is the same equation multiplied by b^T. The momenta are the prognostic variables, Ω is the variable of the solve.

5.3 The EI step

The same in the Verlet structure, EI $= I_{\beta h} \circ K_h \circ E_{(1-\beta)h}$ (Section 6) with the implicit part of the table as H:

- $E_{(1-\beta)h}$: explicit, $\vartheta \leftarrow \vartheta + (1-\beta)h\,[\text{implicit column of } \vartheta]$ (with the old Ω) and $M \leftarrow M - (1-\beta)h\,[b\,G_\eta p + (0, g\rho)^T]$;
- K_h: $U \leftarrow U + h\,G_x \tilde{p}^{\,n+1/2}$ (the extrapolated pressure of the half step);
- $I_{\beta h}$: steps 2–5 above with $(1-\beta)$ replaced by 0 in $\hat{M}$, $\hat{\rho}$, $\hat{\Theta}$ (the old-time parts are already in E), $\hat{M}$ the momenta after K_h, and the horizontal divergence $\beta h\,J^{-1}\partial_x(JU)$ of the U after K_h.

The column solve is the same Ω system.

5.4 Column-locality on the C-grid

On the C-grid U lives at the x-faces, W and Ω at the η-faces of the columns, and b, b^T contain four-point means: $\Omega = W - z_x\bar{U}$, and the metric term of U is $z_x\overline{G_\eta p}$ (mean over the four η-faces around the x-face). Then $b^T b\,G_\eta p = G_\eta p + z_x\overline{z_x\overline{G_\eta p}}$ contains the pressure of the neighbouring columns. Two couplings, treated as follows:

(a) Approximate Jacobian: the right-hand side of the linearly implicit step contains the exact discrete operator at the current state; only the matrix of the Ω system is column-local, with $1 + z_x^2$ of the own column. The remainder $z_x\overline{z_x\overline{G_\eta p}} - z_x^2 G_\eta p$ (a horizontal second difference) enters only through the increment of the step. Step 5 uses the exact discrete b, so U^{n+1} and W^{n+1} are consistent with p^{n+1}; the next step computes $\Omega = W - z_x\bar{U}$ from them.

(b) The horizontal mass flux uses the predictor $\hat{U}$, i.e. U with the metric increment of the implicit step at the old pressure, not U^{n+1}; the difference $\beta h z_x G_\eta(p^{n+1} - p^n)$ would couple the columns through $\partial_x(J z_x G_\eta p^{n+1})$ and enters in the next small step. It is $O(h\,\Delta p)$, so the order is kept. Without the predictor (horizontal flux with U before the metric increment) the lag is $O(h)$ and EI drops to first order with 20 times larger errors (measured).

With (a) and (b) the system is tridiagonal per column, like the Cartesian W system; without them it is two-dimensional ($\mathcal{M} = I - \frac{1}{4}\Delta\tau^2 M^I_{up} M_{pu}$ in the notation of the Imex note). The lag in (b) makes EI on terrain not exactly symmetric and the conjugacy with WRF only approximate; order and stability are to be checked for constant slope.

5.5 Implementation and tests

Both treatments of the metric term are implemented; the switch `FV.FV_METRIC_IMPLICIT[]` (default `false`) selects the implicit one for the frozen `ExnerModel` (Exner pressure or pressure, `pvar = :pi`, `:p`) and `MoistExnerModel` on terrain-following grids; for `FVModel` it is an error. In the example: `METRIC=implicit`.

- *Explicit* (default, as WRF): the metric term is part of the horizontal momentum part; U^{n+1} is known before the column solve, $\Omega^{n+1} = W^{n+1} - z_x\bar{U}^{n+1}$ is an affine shift of W^{n+1}, and the column system is the W system. Sufficient as long as $c_s\Delta\tau|z_x|/(J\Delta\eta) \lesssim 1$ (Agnesi: 0.2).
- *Implicit*: the discrete metric term of $\partial_x p|_z$ at the x-face, $-(z_t c_t - z_b c_b)/(J_x\Delta\eta)$ with c the face means of the two columns, splits exactly into

$$z_t c_t - z_b c_b = \bar{z}_x\,(c_t - c_b) + \bar{c}\,(z_t - z_b),$$

the vertical difference (`mvert`) and a remainder without vertical pressure derivative. `mvert` and the horizontal part $-z_x\bar{U}$ of Ω move from the horizontal parts (`:hmomentum`, `:hthermo`) to the vertical part (`:vertical`); the sum of the parts, i.e. the operator, is unchanged (tested to 10^{-16}). The column solve (`exner_column_solve!`, `moist_column_solve!`) is the Ω system with $1+z_x^2$ and right-hand side $R_W - z_x\bar{R}_U$; after the back substitution `_momenta_from_pressure!` computes U and W from the new pressure. The predictor of (b) is the hook `frozen_hthermo!` (weight βh in EI, h in WRF). The option acts in every fast integrator with a column solve (`:hevi`, `:ei`, `:wrf`) and with both options ε, γ_d.

Order (one frozen stage, hill grid, $T = 8\,\mathrm{s}$, against RK4, implicit metric term): EI 2.0, HEVI 2.0, WRF 1.0, EI with $\varepsilon = 0.1$ 1.0, in the Exner, pressure and moist forms; the errors of EI equal those with the explicit metric term ($8.0 \cdot 10^{-5}$ against $7.3 \cdot 10^{-5}$ with 8 steps).

Steep terrain. Agnesi hill $h_0 = 2\,\mathrm{km}$, $a = 2\,\mathrm{km}$ (max $|z_x| = 0.6$), $\Delta x = 1\,\mathrm{km}$, $\Delta\eta = 100\,\mathrm{m}$, $\Delta\tau = 2.15\,\mathrm{s}$, $c_s\Delta\tau|z_x|/\Delta z = 5.8$; pressure form, frozen fast loop from rest; growth $\|\delta z^n\|/\|\delta z^0\|$ of a perturbation:

metric term	fast integrator	250 steps	1000 steps
explicit	EI, WRF, $\varepsilon = \gamma_d = 0.1$	10^{77}	overflow
explicit	EI, $\varepsilon = 0.1$	$7 \cdot 10^{69}$	$2 \cdot 10^{270}$
implicit	EI, $\varepsilon = \gamma_d = 0.1$	$1.1 \cdot 10^3$	$2.7 \cdot 10^3$
implicit	WRF, $\varepsilon = \gamma_d = 0.1$	$1.1 \cdot 10^3$	$1.2 \cdot 10^3$
implicit	EI, $\varepsilon = 0.1$	$1.1 \cdot 10^3$	$1.4 \cdot 10^3$
implicit	EI, $\varepsilon = 0.05$	$3 \cdot 10^6$	$5 \cdot 10^{17}$

(The bounded value $\approx 10^3$ is the response to the perturbation, not growth: it does not increase.) With the implicit metric term the instability of the explicit one disappears, but off-centring is still needed: without it (EI, HEVI) a slower growth remains (10^{14}–10^{20} after 300 steps), with $\varepsilon = 0.05$ at this slope as well; at $c_s\Delta\tau|z_x|/\Delta z = 2.3$ (grid 32×40) $\varepsilon = 0.05$ suffices. The remaining growth is attributed to the column-local matrix (a): on this hill z_x changes strongly between neighbouring columns. Test (`test/runtests.jl`, "implicit metric term"): operator unchanged, parts sum to the operator, orders, and on the 32×40 steep grid explicit unstable, implicit bounded for EI and WRF with $\varepsilon = \gamma_d = 0.1$.

Agnesi, pressure form (Section 9, $\max|z_x| = 0.026$, so the explicit metric term is sufficient): With EI and $\varepsilon = \gamma_d = 0.1$ the implicit metric term changes the θ' errors by less than 10 %; without options also, except MIS3C at 20 s (0.0016 → 0.0063) and the weakly unstable MIS4a cases at 60–120 s. On a frozen stage the two treatments differ by $O(\Delta\tau^2)$ (on one frozen stage of the Agnesi grid: $6.7 \cdot 10^{-6}$ and $6.5 \cdot 10^{-8}$ in U for 5 and 50 small steps). With WRF the θ' error rises to 0.02–0.046, nearly independent of Δt (the small step $h \approx \Delta\tau_{\mathrm{ac}}$ is the same): implicit and explicit WRF differ at first order in h ($5.2 \cdot 10^{-3}$ and $4.7 \cdot 10^{-4}$ in U on the same stage), concentrated at the lowest levels over the hill. With the implicit metric term the pressure gradient of the U equation is split into the horizontal part $G_x p$ and the metric part $z_x G_\eta p$, each of them large (hydrostatic, $\approx z_x\rho g$) and cancelling in the sum. EI treats both symmetrically (E and I), WRF applies the first in the forward kick and the second β-weighted in the column solve; the resulting first-order imbalance is larger than WRF's own error. Two variants that were tried do not remove it: the predictor weight βh instead of h (makes WRF conjugate to EI, but the change of variables E then carries $(1-\beta)hz_xG_\eta p$, and the stage-boundary defect grows a hundredfold), and the sum $\beta h\, z_x G_\eta(p_{\mathrm{end}} - p_{\mathrm{start}})$ of the per-step differences explains only a tenth of the difference. *Recommendation: the implicit metric term with EI.* For WRF a δ-form (metric force at the old pressure in the kick, only the increment $\beta h z_x G_\eta \delta p$ in the column solve) is an open item.

Table 1: Pressure form, metric term explicit and implicit: relative L^2 error of θ' against Δt (s).

	fast integrator, metric term	10	20	40	60	90	120	180	240
MIS2	EI, explicit	0.003	0.002	0.006	0.005	0.019	0.010	0.033	u
	EI, implicit	0.003	0.002	0.006	0.005	0.019	0.011	0.032	u
	EI $\varepsilon = \gamma_d = 0.1$, explicit	0.001	0.002	0.004	0.004	0.007	0.008	0.010	u
	EI $\varepsilon = \gamma_d = 0.1$, implicit	0.001	0.002	0.004	0.004	0.007	0.008	0.010	u
	WRF $\varepsilon = \gamma_d = 0.1$, explicit	0.002	0.002	0.004	0.004	0.007	0.008	0.010	u
	WRF $\varepsilon = \gamma_d = 0.1$, implicit	0.023	0.038	0.029	0.038	0.037	0.039	0.044	u
MIS3C	EI, explicit	0.002	0.002	0.042	0.029	0.038	0.070	0.055	0.090
	EI, implicit	0.002	0.006	0.041	0.029	0.038	0.071	0.056	0.092
	EI $\varepsilon = \gamma_d = 0.1$, explicit	0.002	0.002	0.002	0.002	0.003	0.003	0.005	0.013
	EI $\varepsilon = \gamma_d = 0.1$, implicit	0.001	0.001	0.002	0.002	0.002	0.003	0.004	0.013
	WRF $\varepsilon = \gamma_d = 0.1$, explicit	0.002	0.002	0.002	0.002	0.003	0.003	0.004	0.013
	WRF $\varepsilon = \gamma_d = 0.1$, implicit	0.020	0.040	0.040	0.041	0.039	0.040	0.042	0.043
MIS4a	EI, explicit	0.001	0.002	0.003	**0.156**	u	**0.102**	u	u
	EI, implicit	0.001	0.002	0.003	**0.132**	u	**0.132**	u	u
	EI $\varepsilon = \gamma_d = 0.1$, explicit	0.001	0.002	0.002	0.002	0.003	0.006	0.005	0.006
	EI $\varepsilon = \gamma_d = 0.1$, implicit	0.001	0.001	0.002	0.002	0.003	0.006	0.004	0.006
	WRF $\varepsilon = \gamma_d = 0.1$, explicit	0.002	0.002	0.002	0.002	0.003	0.006	0.004	0.006
	WRF $\varepsilon = \gamma_d = 0.1$, implicit	0.019	0.034	0.035	0.035	0.042	0.038	0.042	0.042
WSRK3	EI, explicit	0.008	0.032	0.050	0.052	0.073	0.068	**0.102**	**0.111**
	EI, implicit	0.007	0.033	0.051	0.054	0.074	0.069	**0.103**	**0.112**
	EI $\varepsilon = \gamma_d = 0.1$, explicit	0.001	0.002	0.002	0.003	0.006	0.004	0.010	0.016
	EI $\varepsilon = \gamma_d = 0.1$, implicit	0.001	0.002	0.002	0.003	0.006	0.004	0.010	0.016
	WRF $\varepsilon = \gamma_d = 0.1$, explicit	0.002	0.002	0.003	0.004	0.006	0.004	0.010	0.016
	WRF $\varepsilon = \gamma_d = 0.1$, implicit	0.025	0.029	0.040	0.044	0.043	0.044	0.045	0.046

6 Linear algebra

6.1 The column system

In a column i with $\Delta z = J_i \Delta\eta$, cells $k = 1, \dots, n_z$ and faces $k = 1, \dots, n_z + 1$, the vertical part acts only on ϑ and on the vertical momentum (W, with $W_1 = W_{n_z+1} = 0$; on terrain with the explicit metric term the same system holds for W). With frozen coefficients it is linear,

$$J_v = \begin{pmatrix} 0 & D_v \\ B_v & 0 \end{pmatrix} \quad \text{on} \quad \begin{pmatrix} \vartheta \\ W \end{pmatrix}.$$

For every thermodynamic variable ϕ with the frozen face factor ϕ_f (1 for ρ, θ_f for Θ, the face mixing ratios for the moist variables)

$$(D_v W)_{\phi,k} = -\frac{\phi_{f,k+1} W_{k+1} - \phi_{f,k} W_k}{\Delta z},$$

and at the face k

$$(B_v \vartheta)_k = -\frac{\delta p_k}{\Delta z} - \frac{g}{2}(\rho_{k-1} + \rho_k).$$

The linearized face pressure difference is

form	δp_k	coefficients
product	$\alpha_k \Theta_k - \beta_k \Theta_{k-1}$	$\alpha_k = \beta_k = P_{f,k}$
`:gradp`	$\alpha_k \Theta_k - \beta_k \Theta_{k-1}$	$\alpha_k = P_k$, $\beta_k = P_{k-1}$ (cell values at the stage start)
pressure variable	$p_k - p_{k-1}$	$P_k = 1$
Exner, moist	$P_k(\pi_k - \pi_{k-1})$	$P_k = c_p \Theta_{f,k}$

with, for the extended forms, the face factors c_k^B, c_k^T of the pressure equation,

$$(D_v W)_{\pi,k} = -\frac{c_{k+1}^B W_{k+1} - c_k^T W_k}{\Delta z}.$$

Two-factor solve. All schemes need $Y = (I - DJ_v)^{-1}R$ with the row scaling $D = \mathrm{diag}(c_\vartheta I,\ c_w I)$:

$$\begin{pmatrix} I & -c_\vartheta D_v \\ -c_w B_v & I \end{pmatrix} \begin{pmatrix} Y_\vartheta \\ Y_W \end{pmatrix} = \begin{pmatrix} R_\vartheta \\ R_W \end{pmatrix}.$$

The first block row gives

$$Y_\vartheta = R_\vartheta + c_\vartheta D_v Y_W;$$

inserted into the second (Schur complement on the W block):

$$\big(I - c_\vartheta c_w\, B_v D_v\big)\, Y_W = R_W + c_w\, B_v R_\vartheta.$$

With

$$r = \frac{c_\vartheta c_w}{\Delta z^2},$$
$$g_g = \frac{c_\vartheta c_w g}{2\Delta z},$$

row $k = 2, \dots, n_z$ of the product and `:gradp` forms is

$$\big[-r\beta_k\theta_{f,k-1} + g_g\big]Y_{W,k-1} + \big[1 + r\theta_{f,k}(\alpha_k + \beta_k)\big]Y_{W,k} + \big[-r\alpha_k\theta_{f,k+1} - g_g\big]Y_{W,k+1}$$
$$= R_{W,k} - \frac{c_w}{\Delta z}\big(\alpha_k R_{\Theta,k} - \beta_k R_{\Theta,k-1}\big) - \frac{c_w g}{2}\big(R_{\rho,k-1} + R_{\rho,k}\big),$$

and of the pressure, Exner and moist forms

$$\big[-rP_k c_{k-1}^T + g_g\big]Y_{W,k-1} + \big[1 + rP_k(c_k^T + c_k^B)\big]Y_{W,k} + \big[-rP_k c_{k+1}^B - g_g\big]Y_{W,k+1}$$
$$= R_{W,k} - \frac{c_w}{\Delta z}P_k\big(R_{\pi,k} - R_{\pi,k-1}\big) - \frac{c_w g}{2}\big(R_{\rho,k-1} + R_{\rho,k}\big)$$

(the coefficient of $Y_{W,k-1}$ for $k = 2$ and of $Y_{W,k+1}$ for $k = n_z$ vanish). This is the discrete form of the Helmholtz operator of Section 3. The pressure part is a discrete $-c_s^2\partial_{zz}$ with positive diagonal, the gravity part $\pm g_g$ is antisymmetric and small ($g_g/r = g\Delta z/2 \ll c_s^2$); the system is solved by the Thomas algorithm without pivoting, factored once per stage and value of $c_\vartheta c_w$ (`_frozen_factor`, `_exner_factor`, cached under the key $c_\vartheta c_w$). Back substitution, for every thermodynamic variable:

$$Y_{\phi,k} = R_{\phi,k} - \frac{c_\vartheta}{\Delta z}\big(\phi_{f,k+1}Y_{W,k+1} - \phi_{f,k}Y_{W,k}\big).$$

U rows: $Y_U = R_U$. Moisture enters B_v only through ρ and π, so the moist variables are back substitution only. For the Ω system of Section 5 the same algebra holds with $W \to \Omega$, $c_w \to c_w(1 + z_x^2)$ in the pressure part of B_v and $\Omega = 0$ at the ground; U and W then follow from the pressure. The matrix depends on c_ϑ and c_w only through the product:

	c_ϑ	c_w	$c_\vartheta c_w$
HEVI (Crank–Nicolson step of length h)	$h/2$	$h/2$	$h^2/4$
EI(ε), WRF(ε)	βh	βh	$\beta^2 h^2$
EI version III (γ)	$h/2$	γh	$\gamma h^2/2$

For $\varepsilon = 0$ ($\gamma = \frac{1}{2}$) the matrices are identical.

6.2 The steps as compositions

Let H be the implicit part "thermodynamics and vertical", acting on (ϑ, W) with U as a parameter (affine with the forcing),

$$H(\vartheta, U, W) = \big(D_h U + D_v W + f_\vartheta,\ 0,\ B_v\vartheta + f_W\big).$$

For a scalar weight c define the explicit and the implicit step of H,

$$E_c\, z = z + c\, H(z),$$

$$I_c\, z = z^+ \quad \text{with} \quad z^+ = z + c\, H(z^+),$$

and the momentum kick with extrapolation,

$$K_h\, z = z + h\,\big(0,\ G_x\tilde{\vartheta} + f_U,\ 0\big).$$

I_c is one two-factor solve, $z^+ = z + (I - cJ)^{-1}cH(z)$. U is constant within E_c and I_c, so the horizontal divergence D_hU is the same explicit or implicit. The two schemes are

$$\mathrm{WRF}(\varepsilon, \gamma_d) = I_{\beta h} \circ E_{(1-\beta)h} \circ K_h,$$

$$\mathrm{EI}(\varepsilon, \gamma_d) = I_{\beta h} \circ K_h \circ E_{(1-\beta)h}.$$

($I_{\beta h}E_{(1-\beta)h}$ is the β-weighted Crank–Nicolson step of the column with the new U.) Without H EI is position Verlet and WRF forward–backward; without K both are the (off-centred) Crank–Nicolson step.

6.3 Conjugacy

$E_{(1-\beta)h} = I + (1-\beta)hH$ and $I_{\beta h} = (I - \beta hH)^{-1}$ are functions of the same H and commute. For the affine parts, with $Hz = Jz + a$,

$$E_{(1-\beta)h}I_{\beta h}z - I_{\beta h}E_{(1-\beta)h}z = \big[\beta h(I + (1-\beta)hJ) + (1-\beta)h(I - \beta hJ) - hI\big](I - \beta hJ)^{-1}a = 0.$$

Hence, with $E = E_{(1-\beta)h}$,

$$\mathrm{WRF}(\varepsilon, \gamma_d) = E\ \mathrm{EI}(\varepsilon, \gamma_d)\ E^{-1},$$

and over a stage of n small steps

$$\mathrm{EI}^n = E^{-1}\,\mathrm{WRF}^n\,E = I_{\beta h}K_h\,\mathrm{WRF}^{n-1}E.$$

The extrapolation corresponds as well: if the WRF state is Ez^n (z^n the EI state), WRF's ϑ^n is EI's $\vartheta^{n+1/2}$, both extrapolations start without extrapolation, and the kicks are identical. This holds for arbitrary linear operators (terrain, gravity, forcing), not only for Fourier modes. Consequences:

- for equal options the two small steps have the same eigenvalues: the same stability, dispersion and dissipation;
- they differ by the change of variables E: WRF's thermodynamic variables are EI's $\vartheta^{n+1/2}$;
- in a MIS or RK3 stage ϑ, U, W start at the same time level. WRF starts and ends without E: its stage map differs from EI's by $O(\Delta\tau)$ (for $\varepsilon = 0$) — the only difference between the two schemes. For $\varepsilon = 0$ EI is second order and WRF first order (through the stage boundaries); for $\varepsilon > 0$ both are first order.

Verified in the code (frozen loop, hill grid, 10 steps of 2 s, constant forcing): $\|\mathrm{WRF}^n(Ex) - E(\mathrm{EI}^n x)\|/\|x\| \le 3\cdot 10^{-15}$ for $(\varepsilon, \gamma_d) = (0,0)$, $(0.1,0)$, $(0,0.1)$, $(0.1,0.1)$, $(0.2,0.3)$ in the product, Exner, pressure and moist forms, 2–$3\cdot 10^{-6}$ for `:gradp` (linearly implicit), while without E the two differ by 4–$8\cdot 10^{-4}$. On terrain with the Ω system the lag (b) of Section 5 makes E and I commute only approximately.

Version III (γ, only the W rows off-centred): $E_{(h/2,(1-\gamma)h)}$ and $I_{(h/2,\gamma h)}$ have different weights on the thermodynamic and the W rows and do not commute for $\gamma \neq \frac{1}{2}$; it is not conjugate to WRF for any β and damps less (below).

Combined half steps. Let (e_ϑ, e_w) be the weights of E and (i_ϑ, i_w) those of I (EI(ε): $e_\vartheta = e_w = (1-\beta)h$, $i_\vartheta = i_w = \beta h$; version III: $(\frac{h}{2}, (1-\gamma)h)$ and $(\frac{h}{2}, \gamma h)$). I adds $z^+ - z_s$, z_s the state before I; the next E needs the same increment scaled by e/i. Between two small steps therefore

$$z \leftarrow z^+ + \frac{e_\vartheta}{i_\vartheta}\,(z^+ - z_s) + D_{(0,\ e_w - e_\vartheta i_w/i_\vartheta)}\,G_v(z^+).$$

The last term vanishes for EI(ε) and is $(1-2\gamma)hB_v\vartheta^+$ for version III; $e_\vartheta/i_\vartheta = (1-\beta)/\beta$, i.e. $2z^+ - z_s$ for $\varepsilon = 0$. In terms of the compositions this is $\mathrm{EI}^n = I_{\beta h} K_h\,\mathrm{WRF}^{n-1} E$: EI with combined half steps is WRF, started and finished with the change of variables.

6.4 Fourier analysis

Acoustic model with one pressure-like variable ϑ:

$$\begin{aligned}\vartheta' &= -\mathrm{i}\,c_s\,(k_x u + k_z w),\\ u' &= -\mathrm{i}\,c_s k_x\,\vartheta,\\ w' &= -\mathrm{i}\,c_s k_z\,\vartheta.\end{aligned}$$

With $a_x = c_s k_x h$, $a_z = c_s k_z h$ and

$$\begin{aligned}\mu &= (1-\beta) + \beta\lambda,\\ \nu &= (1-\gamma) + \gamma\lambda,\end{aligned}$$

the ansatz $z^n = \lambda^n z$ gives, besides the neutral mode $\lambda = 1$, the characteristic equations

$$\mathrm{WRF}(\varepsilon),\ \mathrm{EI}(\varepsilon): \quad (\lambda-1)^2 + a_x^2\lambda + a_z^2\,\mu^2 = 0,$$

$$\text{EI version III}(\gamma): \quad (\lambda-1)^2 + a_x^2\lambda + \tfrac{a_z^2}{2}\,\nu\,(1+\lambda) = 0.$$

For $\varepsilon = 0$, $\gamma = \frac{1}{2}$ both are $(\lambda-1)^2 + a_x^2\lambda + a_z^2(1+\lambda)^2/4 = 0$. Horizontally ($a_z = 0$) the schemes are stable for $a_x \le 2$, vertically unconditionally. For $a_x = 0$ the modulus of the damped pair is

$$\mathrm{WRF},\ \mathrm{EI}(\varepsilon): \quad |\lambda|^2 = \frac{1 + (1-\beta)^2 a_z^2}{1 + \beta^2 a_z^2},$$

$$\text{version III}: \quad |\lambda|^2 = \frac{1 + (1-\gamma)a_z^2/2}{1 + \gamma a_z^2/2}.$$

For $a_z \to \infty$: $(1-\beta)^2/\beta^2$ against $(1-\gamma)/\gamma$; $\varepsilon = 0.1$ gives $|\lambda| \to 0.82$, version III with $\gamma = 0.55$ only 0.90 (the same limit needs $\gamma \approx 0.60$). Spectral radius of the non-neutral modes:

a_x	a_z	$\varepsilon = 0$	WRF, EI $\varepsilon = 0.1$	III $\gamma = 0.55$	III $\gamma = 0.6$
0.0	1.0	1.0000	0.9608	0.9802	0.9608
0.0	3.0	1.0000	0.8708	0.9330	0.8699
0.0	20.0	1.0000	0.8198	0.9054	0.8182
1.0	0.3	1.0000	0.9956	0.9978	0.9956
1.5	3.0	1.0000	0.8708	0.9330	0.8699
1.5	20.0	1.0000	0.8198	0.9054	0.9512

7 Comparison

	Verlet EI	WRF forward–backward
horizontal integrator	Störmer–Verlet ($\varepsilon = 0$)	forward–backward (conjugate)
order	2 ($\varepsilon = 0$), 1 ($\varepsilon > 0$)	1
vertical integrator	β-weighted in ϑ and W (version III: W only)	β-weighted in ϑ and W
options	off-centring ε, extrapolation γ_d	off-centring ε, extrapolation γ_d
unknown of the column solve	W; on terrain Ω (Section 5)	W (WRF: after eliminating ϕ)
metric term	explicit (code) or implicit with Ω	explicit
cost per small step	each part once, 1 solve (half steps combined)	each part once, 1 solve
stage boundaries (MIS, RK3)	no defect	$O(\Delta\tau)$ conjugation defect

WRF needs its damping: RK3 has $|g_0| = 1$ for the acoustic modes, and the forward–backward step adds none; divergence damping and off-centring provide it (Wicker and Skamarock 2002; Baldauf 2010). The same options act identically in EI.

8 Implementation in the frozen split

Both schemes are fast integrators of the frozen MIS split, implemented in `frozen_fast_solve!` (`TimeStepping/frozen.jl`) with `fast = :ei` or `fast = :wrf`. They work for the three frozen models `FVModel` (full pressure `:gradp` or product form), `ExnerModel` and `MoistExnerModel`. In the frozen fast system $Z' = F_f(C, Z) + F$ the model splits F_f into the parts

`:hthermo`	H_ϑ: $D_h U$ (incl. $-z_x \bar{U}$ in Ω), forcing of ϑ and W, sponge
`:hmomentum`	H_U: horizontal pressure gradient $G_x p$ (incl. the metric term), forcing of U
`:vertical`	G_v: $D_v W$ in ϑ, $G_z p - g\bar{\rho}$ in W

and provides the column solve $y = (I - DJ_v)^{-1} R$, J_v the Jacobian of G_v (exact for the linear vertical parts of the product, Exner and moist forms, the linearly implicit approximation for `:gradp`), D the row scaling c_ϑ on ϑ and c_w on W; the W system depends on $c_\vartheta c_w$ only and is factored once per stage (`frozen_stage_begin!`). Hooks (`FV/frozen_ei.jl`): `frozen_vertical_rhs!` (DG_v), `frozen_vertical_implicit!` ($Z \leftarrow Z + (I - DJ_v)^{-1} DG_v(Z)$) and `frozen_vertical_solve!` (general right-hand side).

Options. `frozen_fast_solve!`, `frozen_mis_step!` (MIS, Lawson, ETD) and `run_frozen` take the keywords `offcenter` $= \varepsilon$ and `smdiv` $= \gamma_d$ (default 0) for `:ei` and `:wrf` alike; `EI_GAMMA[]` (default $\frac{1}{2}$) selects version III of EI ($\varepsilon = 0$ only); `EI_MERGE[]` (default true) the combined half steps. The weights $(e_\vartheta, e_w, i_\vartheta, i_w)$ are `ei_weights(h, ε)`.

Verlet EI (`:ei`):

$$
\begin{aligned}
&Z \leftarrow Z + e_\vartheta\, H_\vartheta(Z) + D_{(e_\vartheta, e_w)} G_v(Z) && E:\ \vartheta^{n+1/2},\ W \text{ explicit part}\\
&Z \leftarrow Z + h\, H_U\big(Z + \gamma_d (Z - Z_{\mathrm{prev}})\big) && K:\ U^{n+1}\\
&Z \leftarrow Z + i_\vartheta\, H_\vartheta(Z) && I:\ \text{horizontal part}\\
&Z \leftarrow Z + (I - D_{(i_\vartheta, i_w)} J_v)^{-1} D_{(i_\vartheta, i_w)} G_v(Z) && I:\ \vartheta^{n+1},\ W^{n+1};
\end{aligned}
$$

Z_{prev} the state of the previous kick, $Z_{\mathrm{prev}} = Z$ after the first E. For linear G_v and $\varepsilon = 0$ this is exactly (6)–(10). Combined half steps as in Section 6.3: one horizontal momentum, one horizontal thermodynamic and one vertical evaluation and one column solve per small step — as WRF; only

the first E of a stage is extra. For `:gradp` (linearly implicit) and in the sponge (evaluated at the state before I) combined and separate half steps differ by $O(\Delta\tau^2)$ per step ($\le 3\cdot 10^{-6}$ relative in the test), and both are of order 2 for $\varepsilon = 0$.

WRF (`:wrf`):

$$\begin{aligned} Z &\leftarrow Z + h\,H_U\big(Z + \gamma_d(Z - Z_{\mathrm{prev}})\big) && K:\ U^{n+1} \\ Z &\leftarrow Z + (I - \beta h J_v)^{-1} h\big[H_\vartheta(Z) + G_v(Z)\big] && I_{\beta h}E_{(1-\beta)h}:\ \vartheta^{n+1},\ W^{n+1}, \end{aligned}$$

$Z_{\mathrm{prev}} = Z$ at the start of the stage. For linear G_v the second line is (3)–(5).

Tests (`test/runtests.jl`, "Verlet EI and WRF fast integrators"): the two-factor column solve is exact for the linear forms; on one frozen stage of the hill problem (product, `:gradp`, Exner, pressure, moist) EI is of order 2 and WRF of order 1 against RK4, both of order 1 with $\varepsilon = 0.1$; $\mathrm{WRF}^n E = E\,\mathrm{EI}^n$ for $(\varepsilon, \gamma_d) = (0,0), (0.1,0), (0.1,0.1)$; on a flat, horizontally uniform column EI and WRF with $\varepsilon = 0$ coincide with the Crank–Nicolson step of `:hevi` to round-off, and EI stays bounded at the vertical Courant number 35; in MIS2 both conserve mass to round-off on terrain, EI differs from `:hevi` by $O(\Delta\tau^2)$, WRF by $O(\Delta\tau)$; with $\gamma = 0.6$ (version III) EI is of order 1. Combined and separate half steps of EI agree to round-off for the linear forms ($\gamma = 0.5$ and 0.6); no small step is longer than $\Delta t/n_{\mathrm{sub}}$ (Section 9). Full suite: 571 tests pass.

error on one frozen stage ($T = 8$ s, max norm)	8 steps	16 steps	32 steps
EI	0.0242	0.0060	0.0015
WRF, $\beta = 0.5$	0.116	0.055	0.027
WRF, $\beta = 0.55$	0.117	0.057	0.028
WRF, $\beta = 0.55$, $\gamma_d = 0.1$	0.102	0.050	0.025

(Exner form; the other forms agree to two digits.) The linearized path (`fast_solve!`, `:ei`, `:wrf` in `TimeStepping/fast.jl`) has the same two loops for the linear fast operator.

9 Agnesi mountain wave

Set-up as in the MIS study (`docs/mis_agnesi`): hydrostatic wave over the Agnesi hill ($a = 10$ km, $h_0 = 400$ m, $U = 10$ m/s, $N = 0.01\,\mathrm{s}^{-1}$), 200×78 cells, $\Delta x = 2$ km, $\Delta\eta = 200$ m, $T = 10800$ s, $n_{\mathrm{sub}} = \lceil \Delta t/\Delta\tau_{\mathrm{ac}} \rceil$ with $\Delta\tau_{\mathrm{ac}} = 4.3$ s (3 small steps at $\Delta t = 10$ s, 56 at 240 s); reference RK4 of the same semi-discrete operator; errors are relative L^2 errors in $|x - x_m| \le 30$ km, $z \le 9$ km. Two forms of the extended system (`ExnerModel`) with an additional pressure variable: the *Exner pressure* π (W row $-c_p\Theta_f\delta\pi/\Delta z - g\bar\rho$) and the *pressure* p (W row $-\delta p/\Delta z - g\bar\rho$, `pvar = :p`); each column is in discrete hydrostatic balance of its form (rest-state residual of the W rows $4\cdot 10^{-13}$). HEVI is the fast integrator of the MIS study (horizontal Verlet half step, Crank–Nicolson vertical step, horizontal half step). EI and WRF are run with the same options: none, $\varepsilon = 0.1$, and $\varepsilon = \gamma_d = 0.1$ (WRF's standard values).

`FORM=exner|pres FAST=ei|wrf|hevi [OFFCENTER=`ε`] [SMDIV=`γ_d`] julia examples/frozen_agnesi.jl`

Number of small steps per stage (corrected). Stage i of a MIS step integrates over $d_i\Delta t$. The number of small steps was $\max(1, \mathrm{round}(d_i n_{\mathrm{sub}}))$, which allows small steps up to $1.5\,\Delta t/n_{\mathrm{sub}}$, i.e. above the horizontal acoustic limit for which n_{sub} was chosen. MIS2 ($d = (0.127, 0.590, 0.470)$) with $n_{\mathrm{sub}} = 3$ at $\Delta t = 10$ s ran its third stage in one small step of $4.7\,\mathrm{s} > \Delta\tau_{\mathrm{ac}} = 4.3$ s, and with $n_{\mathrm{sub}} = 5$ at 20 s in two steps of 4.7 s: exactly the two cases in which MIS2 with EI and WRF was

Table 2: Exner pressure: relative L^2 error of θ' against Δt (s). u: unstable; bold: error ≥ 0.1.

	fast integrator	10	20	40	60	90	120	180	240
MIS2	HEVI	0.007	0.015	0.015	0.018	0.034	0.038	**0.132**	u
	EI	0.002	0.002	0.004	0.005	0.026	0.034	**0.126**	u
	WRF	0.003	0.002	0.004	0.005	0.030	0.044	**0.315**	u
	EI, $\varepsilon = 0.1$	0.002	0.002	0.003	0.004	0.011	0.031	**0.164**	u
	WRF, $\varepsilon = 0.1$	0.002	0.002	0.003	0.003	0.014	0.038	**0.267**	u
	EI, $\varepsilon = \gamma_d = 0.1$	0.001	0.002	0.003	0.004	0.009	0.017	0.036	u
	WRF, $\varepsilon = \gamma_d = 0.1$	0.002	0.002	0.003	0.003	0.010	0.020	0.046	u
MIS3C	HEVI	0.005	0.018	0.035	u	u	u	u	u
	EI	0.002	0.002	0.031	u	u	u	u	u
	WRF	0.002	0.003	0.029	u	u	u	u	u
	EI, $\varepsilon = 0.1$	0.002	0.002	0.002	0.004	0.006	**0.846**	u	u
	WRF, $\varepsilon = 0.1$	0.002	0.002	0.003	0.005	0.007	**1.340**	u	u
	EI, $\varepsilon = \gamma_d = 0.1$	0.002	0.002	0.002	0.004	0.005	0.008	**0.176**	**0.261**
	WRF, $\varepsilon = \gamma_d = 0.1$	0.002	0.002	0.002	0.004	0.006	0.008	**0.196**	**0.289**
MIS4a	HEVI	0.005	0.015	0.015	0.090	u	**0.153**	u	u
	EI	0.001	0.002	0.004	**0.141**	u	**1.050**	u	u
	WRF	**1.186**	0.002	0.004	u	u	u	u	u
	EI, $\varepsilon = 0.1$	0.002	0.002	0.003	0.005	0.011	0.010	0.026	**2.037**
	WRF, $\varepsilon = 0.1$	**0.413**	0.002	0.003	0.005	0.011	0.010	0.030	u
	EI, $\varepsilon = \gamma_d = 0.1$	0.001	0.002	0.003	0.004	0.009	0.009	0.021	0.032
	WRF, $\varepsilon = \gamma_d = 0.1$	0.002	0.002	0.003	0.005	0.010	0.009	0.023	0.032
WSRK3	HEVI	u	u	u	u	u	u	u	u
	EI	0.013	u	u	u	u	u	u	u
	WRF	u	u	u	u	u	u	u	u
	EI, $\varepsilon = 0.1$	0.002	0.002	u	u	**0.135**	u	u	**0.672**
	WRF, $\varepsilon = 0.1$	0.002	0.002	u	u	**0.191**	u	u	**0.719**
	EI, $\varepsilon = \gamma_d = 0.1$	0.001	0.002	u	u	0.013	0.069	**0.131**	**0.172**
	WRF, $\varepsilon = \gamma_d = 0.1$	0.002	0.002	u	u	0.013	0.080	**0.145**	**0.183**

unstable in the first version of this note. HEVI survived them, because its horizontal Verlet steps are half steps. The count is now

$$n_i = \max\left(1, \lceil d_i n_{\text{sub}} \rceil\right) \qquad (\texttt{fast_substeps}),$$

so that no small step is longer than $\Delta t / n_{\text{sub}}$, for MIS, ETD, Lawson and EPIRK and in the frozen and the linearized split. The MIS study of `docs/mis_agnesi` used the rounded count; its HEVI numbers change slightly (e.g. MIS2 at 10 s: θ' error 0.014 $\to$ 0.007). All results below use the corrected count.

EI and WRF with the same options behave the same. In both forms and for all three option sets the stable and unstable cases of EI and WRF coincide, and the errors agree to one or two digits, EI's being slightly smaller (no change of variables missing at the stage boundaries). The few differences are at the edges, where the stage-boundary defect of WRF matters: MIS4a at 10 s ($n_{\text{sub}} = 3$, first stage one step) is weakly unstable with WRF for $\gamma_d = 0$ (Exner: θ' error 1.19 and 0.41; pressure: 0.50 and 0.18) and fine with EI; WSRK3 at 10 and 20 s without options (w error 7.2 and 4.6 with WRF against 0.17 and 0.15 with EI, pressure form).

Accuracy. With $\varepsilon = 0.1$, EI and WRF reduce the θ' error at $\Delta t \leq 60$ s from 0.5–4.5 $\cdot 10^{-2}$ (HEVI) to 1.4–5 $\cdot 10^{-3}$, for all four methods and both forms (without options at $\Delta t \leq 20$ s): with HEVI the θ' error is dominated by the horizontal–vertical splitting of the fast step (the Strang splitting of the stiff vertical part), not by the MIS method. EI and WRF use $D_h U^{n+1}$ and $D_v W^{n+1}$ in the same thermodynamic update and avoid it. The w error (≈ 0.09) is the same for all stable runs: the undamped vertical acoustic modes of the MIS study.

Table 3: Pressure as additional variable: relative L^2 error of θ' against Δt (s).

	fast integrator	10	20	40	60	90	120	180	240
MIS2	HEVI	0.008	0.015	0.014	0.019	0.023	0.018	0.040	u
	EI	0.003	0.002	0.006	0.005	0.019	0.010	0.033	u
	WRF	0.003	0.003	0.006	0.005	0.019	0.011	0.032	u
	EI, $\varepsilon = 0.1$	0.002	0.002	0.004	0.004	0.008	0.008	0.011	u
	WRF, $\varepsilon = 0.1$	0.002	0.002	0.004	0.004	0.007	0.009	0.011	u
	EI, $\varepsilon = \gamma_d = 0.1$	0.001	0.002	0.004	0.004	0.007	0.008	0.010	u
	WRF, $\varepsilon = \gamma_d = 0.1$	0.002	0.002	0.004	0.004	0.007	0.008	0.010	u
MIS3C	HEVI	0.005	0.018	0.045	0.027	0.032	0.061	0.058	u
	EI	0.002	0.002	0.042	0.029	0.038	0.070	0.055	0.090
	WRF	0.002	0.002	0.039	0.029	0.037	0.068	0.054	0.088
	EI, $\varepsilon = 0.1$	0.002	0.002	0.002	0.003	0.003	0.004	0.006	0.013
	WRF, $\varepsilon = 0.1$	0.002	0.002	0.002	0.003	0.003	0.004	0.006	0.013
	EI, $\varepsilon = \gamma_d = 0.1$	0.002	0.002	0.002	0.002	0.003	0.003	0.005	0.013
	WRF, $\varepsilon = \gamma_d = 0.1$	0.002	0.002	0.002	0.002	0.003	0.003	0.004	0.013
MIS4a	HEVI	0.005	0.015	0.015	0.089	u	0.032	u	u
	EI	0.001	0.002	0.003	**0.156**	u	**0.102**	u	u
	WRF	**0.500**	0.002	0.004	u	u	u	u	u
	EI, $\varepsilon = 0.1$	0.002	0.002	0.002	0.003	0.003	0.006	0.005	0.007
	WRF, $\varepsilon = 0.1$	**0.184**	0.002	0.003	0.003	0.004	0.007	0.005	0.007
	EI, $\varepsilon = \gamma_d = 0.1$	0.001	0.002	0.002	0.002	0.003	0.006	0.005	0.006
	WRF, $\varepsilon = \gamma_d = 0.1$	0.002	0.002	0.002	0.002	0.003	0.006	0.004	0.006
WSRK3	HEVI	u	u	0.040	0.047	0.059	0.054	0.090	0.099
	EI	0.008	0.032	0.050	0.052	0.073	0.068	**0.102**	**0.111**
	WRF	0.021	0.052	0.050	0.052	0.077	0.068	**0.104**	**0.113**
	EI, $\varepsilon = 0.1$	0.002	0.002	0.002	0.004	0.007	0.005	0.015	0.016
	WRF, $\varepsilon = 0.1$	0.002	0.002	0.003	0.004	0.007	0.005	0.015	0.017
	EI, $\varepsilon = \gamma_d = 0.1$	0.001	0.002	0.002	0.003	0.006	0.004	0.010	0.016
	WRF, $\varepsilon = \gamma_d = 0.1$	0.002	0.002	0.003	0.004	0.006	0.004	0.010	0.016

Off-centring and extrapolation.

- Without options EI, WRF and HEVI have irregular stability gaps in the Exner form (MIS3C from 60 s, MIS4a at 60–120 s, WSRK3 almost everywhere); in the pressure form MIS2 (up to 180 s), MIS3C and WSRK3 are stable (HEVI: WSRK3 unstable at 10–20 s), MIS4a still has the gap at 60–90 s and from 180 s (WRF from 60 s).
- $\varepsilon = 0.1$ removes the gaps at small and medium Δt and reduces the errors at $\Delta t \geq 40$ s by a factor 2–20. In the pressure form all four methods are then stable up to 240 s (MIS2 up to 180 s), WSRK3 included, with θ' errors ≤ 0.017.
- $\gamma_d = 0.1$ matters only in the Exner form at $\Delta t \geq 120$ s (MIS3C, MIS4a and WSRK3 stable up to 240 s except WSRK3 at 40–60 s); in the pressure form it changes the errors by a few per cent.

Pressure against Exner pressure. With the pressure as additional variable all methods are markedly more robust: with $\varepsilon = 0.1$, MIS3C at 240 s has θ' error 0.013 (Exner: unstable; 0.26 with $\gamma_d = 0.1$), MIS4a 0.0065 (Exner: 2.0 or unstable; 0.032 with γ_d), WSRK3 0.016 (Exner with γ_d: unstable at 40–60 s, 0.17 at 240 s), and HEVI makes WSRK3 stable from 40 s. A plausible cause, not analysed here: in the pressure form the W row $-\delta p$ has no frozen coefficient, while the Exner form carries the frozen $c_p\Theta_f$ in front of $\delta\pi$ and in the π equation, so that the frozen fast operator of a stage and the slow remainder differ more (the constraint-defect modes of the MIS study).

Table 4: Exner pressure: relative L^2 error of w. Bold: error ≥ 1.

	fast integrator	10	20	40	60	90	120	180	240
MIS2	HEVI	0.056	0.104	0.100	0.107	0.126	0.205	**1.173**	u
	EI	0.054	0.098	0.093	0.097	0.125	0.203	**1.211**	u
	WRF	0.068	0.100	0.095	0.099	0.147	0.262	**2.503**	u
	EI, $\varepsilon = 0.1$	0.089	0.095	0.093	0.096	0.092	0.191	**1.341**	u
	WRF, $\varepsilon = 0.1$	0.089	0.096	0.094	0.098	0.106	0.237	**2.088**	u
	EI, $\varepsilon = \gamma_d = 0.1$	0.091	0.092	0.092	0.095	0.087	0.154	0.350	u
	WRF, $\varepsilon = \gamma_d = 0.1$	0.092	0.093	0.094	0.096	0.090	0.172	0.408	u
MIS3C	HEVI	0.186	0.139	0.139	u	u	u	u	u
	EI	0.186	0.136	0.134	u	u	u	u	u
	WRF	0.193	0.150	0.121	u	u	u	u	u
	EI, $\varepsilon = 0.1$	0.089	0.090	0.088	0.164	0.087	**6.007**	u	u
	WRF, $\varepsilon = 0.1$	0.089	0.091	0.089	0.180	0.089	**9.494**	u	u
	EI, $\varepsilon = \gamma_d = 0.1$	0.091	0.091	0.090	0.102	0.087	0.106	**1.133**	**1.800**
	WRF, $\varepsilon = \gamma_d = 0.1$	0.092	0.092	0.090	0.104	0.088	0.107	**1.256**	**1.972**
MIS4a	HEVI	0.154	0.093	0.097	**7.068**	u	**1.892**	u	u
	EI	0.151	0.087	0.090	**8.905**	u	**9.743**	u	u
	WRF	**1.922**	0.089	0.090	u	u	u	u	u
	EI, $\varepsilon = 0.1$	0.090	0.091	0.091	0.095	0.150	0.092	0.173	**14.300**
	WRF, $\varepsilon = 0.1$	**8.840**	0.093	0.092	0.097	0.152	0.092	0.191	u
	EI, $\varepsilon = \gamma_d = 0.1$	0.091	0.091	0.091	0.096	0.145	0.086	0.151	0.085
	WRF, $\varepsilon = \gamma_d = 0.1$	0.092	0.092	0.091	0.096	0.146	0.086	0.154	0.090
WSRK3	HEVI	u	u	u	u	u	u	u	u
	EI	**3.592**	u	u	u	u	u	u	u
	WRF	u	u	u	u	u	u	u	u
	EI, $\varepsilon = 0.1$	0.093	0.092	u	u	0.934	u	u	**2.555**
	WRF, $\varepsilon = 0.1$	0.094	0.093	u	u	**1.318**	u	u	**2.780**
	EI, $\varepsilon = \gamma_d = 0.1$	0.091	0.091	u	u	0.178	**2.278**	0.522	0.589
	WRF, $\varepsilon = \gamma_d = 0.1$	0.092	0.092	u	u	0.185	**2.526**	0.620	0.631

Cost. Serial runs, MIS4a at $\Delta t = 60\,\mathrm{s}$ ($n_{\mathrm{sub}} = 14$, $T = 10800\,\mathrm{s}$, the first Δt of each run excluded because of compilation; Exner form): HEVI 89 s, EI (combined half steps, version III $\gamma = 0.55$) 49 s, WRF ($\varepsilon = \gamma_d = 0.1$) 44 s. The combined EI costs 10 % more than WRF (the first E of each stage and, for version III, the column-local correction) and 55 % of HEVI.

10 Open decisions

1. Default fast integrator: EI or WRF, both with $\varepsilon = 0.1$ (and $\gamma_d = 0.1$ for the Exner form). They are conjugate; EI avoids the stage-boundary defect at the same cost.

2. Default pressure variable of the extended system: p is markedly more robust than π on Agnesi; to be checked on the moist cases (`MoistExnerModel`, `pvar = :p`).

3. The remaining gaps (Exner form; MIS4a without options): linear stability analysis of the coupled MIS–EI/WRF step.

4. Rerun of the MIS study (`docs/mis_agnesi`) with the corrected number of small steps.

5. Orography: the implicit metric term (`FV_METRIC_IMPLICIT`, Section 5.5) is implemented and works with EI; full MIS runs on steep terrain (beyond $c_s \Delta\tau |z_x| / \Delta z \approx 1$) are still to be done; the remaining growth without off-centring on steep terrain (column-local matrix (a)); WRF with the implicit metric term needs a δ-form.

Table 5: Pressure as additional variable: relative L^2 error of w.

	fast integrator	10	20	40	60	90	120	180	240
MIS2	HEVI	0.074	0.099	0.096	0.104	0.113	0.111	0.108	u
	EI	0.077	0.091	0.091	0.095	0.105	0.103	0.099	u
	WRF	0.094	0.092	0.093	0.097	0.107	0.105	0.102	u
	EI, $\varepsilon = 0.1$	0.088	0.095	0.091	0.095	0.098	0.102	0.098	u
	WRF, $\varepsilon = 0.1$	0.088	0.095	0.092	0.097	0.100	0.104	0.101	u
	EI, $\varepsilon = \gamma_d = 0.1$	0.091	0.091	0.091	0.093	0.097	0.101	0.096	u
	WRF, $\varepsilon = \gamma_d = 0.1$	0.091	0.092	0.093	0.095	0.099	0.103	0.099	u
MIS3C	HEVI	0.189	0.156	0.122	0.104	0.112	0.130	0.148	u
	EI	0.187	0.149	0.110	0.096	0.102	0.125	0.117	0.141
	WRF	0.195	0.166	0.110	0.098	0.103	0.123	0.116	0.137
	EI, $\varepsilon = 0.1$	0.089	0.089	0.093	0.089	0.097	0.092	0.093	0.086
	WRF, $\varepsilon = 0.1$	0.089	0.089	0.094	0.090	0.098	0.093	0.093	0.087
	EI, $\varepsilon = \gamma_d = 0.1$	0.091	0.091	0.091	0.090	0.095	0.091	0.091	0.087
	WRF, $\varepsilon = \gamma_d = 0.1$	0.092	0.092	0.092	0.091	0.095	0.092	0.091	0.088
MIS4a	HEVI	0.141	0.106	0.097	**5.841**	u	0.883	u	u
	EI	0.137	0.102	0.091	**9.667**	u	**1.184**	u	u
	WRF	**11.220**	0.106	0.092	u	u	u	u	u
	EI, $\varepsilon = 0.1$	0.090	0.092	0.091	0.089	0.095	0.098	0.086	0.094
	WRF, $\varepsilon = 0.1$	**7.884**	0.093	0.092	0.090	0.096	0.100	0.086	0.095
	EI, $\varepsilon = \gamma_d = 0.1$	0.091	0.091	0.090	0.090	0.094	0.097	0.087	0.093
	WRF, $\varepsilon = \gamma_d = 0.1$	0.092	0.092	0.091	0.090	0.096	0.099	0.087	0.094
WSRK3	HEVI	u	u	0.143	0.159	0.204	0.128	0.133	0.162
	EI	0.174	0.149	0.146	0.139	0.223	0.098	0.154	0.140
	WRF	**7.230**	**4.551**	0.144	0.145	0.234	0.114	0.159	0.141
	EI, $\varepsilon = 0.1$	0.093	0.094	0.089	0.087	0.087	0.080	0.092	0.080
	WRF, $\varepsilon = 0.1$	0.094	0.095	0.090	0.088	0.087	0.080	0.092	0.077
	EI, $\varepsilon = \gamma_d = 0.1$	0.091	0.090	0.089	0.087	0.082	0.082	0.085	0.083
	WRF, $\varepsilon = \gamma_d = 0.1$	0.092	0.091	0.090	0.089	0.082	0.081	0.085	0.082

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